

The Operational Question

Why Did Volumetric Flow Rates Fluctuate?

Core Challenge: Unexplained output dips were detected across operating plant sectors (Zone West, Central, East, North, South).

External Hypothesis: Did external rainfall or outdoor temperatures suppress fluid throughput?

Internal Hypothesis: Or were variances driven by operating pressure instability and technician headcount allocation?

Objective: Integrate 3 disparate data streams into a unified Master DataFrame to perform statistical correlation analysis.



Multi-Source Data Integration

SOURCE 1: INTERNAL



Operational csv

Cleaned sensor logs (cleaned_sensor_readings.csv). Grouped by date and zone to aggregate daily mean **Pressure (PSI)**, mean **Temp (°C)**, and total **Flow Rate (LPM)**.

SOURCE 2: EXTERNAL



Weather REST API

Fetches real-time **Precipitation (mm)** and **Max Outdoor Temp (°C)** using requests.get() wrapped in try-except exception handling to guarantee fallback reliability.

SOURCE 3: DATABASE



SQLite Metadata

Queried relational zones and shifts tables using SQL JOIN and GROUP BY to aggregate total **Assigned Staff** and **Target Flow Rates** directly into Pandas.

Insights & Business Action

Key Pipeline Findings

Zero Weather Correlation:

External rainfall showed negligible correlation () with volumetric flow rates.

Pressure Dominance: Internal operating pressure () is the primary driver of daily flow throughput.

Staffing Impact: Zones with full technician allocations () consistently met target flow capacity.

Business Recommendations

Stabilize Pressure Baselines: Focus engineering controls on maintaining baseline pressure at ~200 PSI rather than monitoring external weather.

Dynamic Shift Scheduling: Align technician staffing headcounts dynamically with zone capacity ratings (High vs Medium capacity).

Automate Data Pipeline: Operationalize this multi-source ETL script into production for automated daily shift reporting.